

Engineering Portfolio Project

Formula 1 CAD Modelling and CFD Analysis

Aerodynamic Investigation of a DRS Rear Wing

Stefano Tamo

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1. Study Objective

1.1 Objectives

The objectives of this study are on one side to assess the aerodynamic performances of a Formula 1 car spoiler with two different DRS configurations (open and closed), and to analyse how the aerodynamic forces are modified when the DRS configuration changes. Finally, the drag coefficient C_d was computed and analysed for both configurations.

1.2 Methodology

A whole Formula 1 car CAD model has been developed. The spoiler of this model has been used for this study. First, simulations with a wind velocity of 10 m/s were run to validate the settings, both on the DRS open and DRS closed configurations, and then the same type of simulations were run this time with a wind velocity of 50 m/s, which is closer from the real speeds at which Formula 1 cars usually race.

2. Formula 1 CAD Modelling

2.1 CAD Model Overview

In order to model the Formula 1 car, pictures of a real car have been used. These pictures can be found in [1]. More precisely, the car is the Ferrari SF24 evo F1 Formula 1 (2024). The scale has been established with the software ImageJ, as well as all the required measurements. The different parts of the car have been modeled separately, namely

- The two front wheels
- The two rear wheels
- The car body
- The front wing
- The rear wing (spoiler)

Then all of these separate parts were assembled in order to obtain the complete car.

2.2 Modelling Software

The software that was used is 3DEXPERIENCE.

2.3 General Vehicle Geometry

A geometry of the Formula 1 model is displayed on picture 1. On this picture the basic dimensions of the car are displayed.

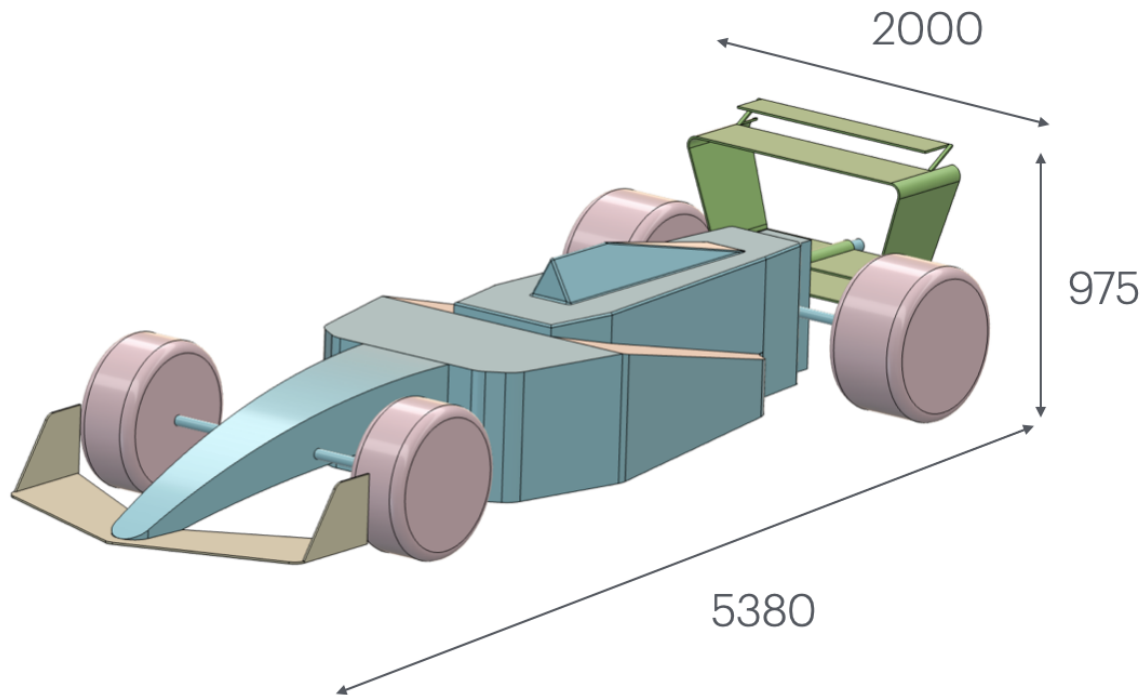


Figure 1: Car geometry and dimensions (in *mm*)

2.4 Rear Wing Geometry

The rear wing geometry has been of course simplified compared to the geometry of a real Formula 1 car spoiler, the goal being here to have a geometry that could be quickly available for the CFD simulations. Two different geometries were created: one where the drag reduction system (DRS) is closed, and one where it is open. These two geometries can be found on pictures 2a and 2b. In order to have a reference regarding the spoiler geometry, the picture found in [2] was used.

2.5 Drag Reduction System

The drag reduction system is a device that can have two configurations: one closed configuration, and one open configuration. While it is closed, the relative movement of air around it creates a downward aerodynamic force that sticks the car to the ground. While it is open, this force decreases in magnitude, and the induced drag is decreased. Although the car reaches higher speed because of the lower drag, it is a bit less stable, as the downward force is reduced. Physically speaking, the DRS device is a flap whose angle can be tuned from 0 to approximately 45 degrees, with 0 degree representing the closed configuration, and 45 degrees the open configuration.

2.5.1 DRS configurations

On figure 2, the two geometries that were used for this study are displayed.

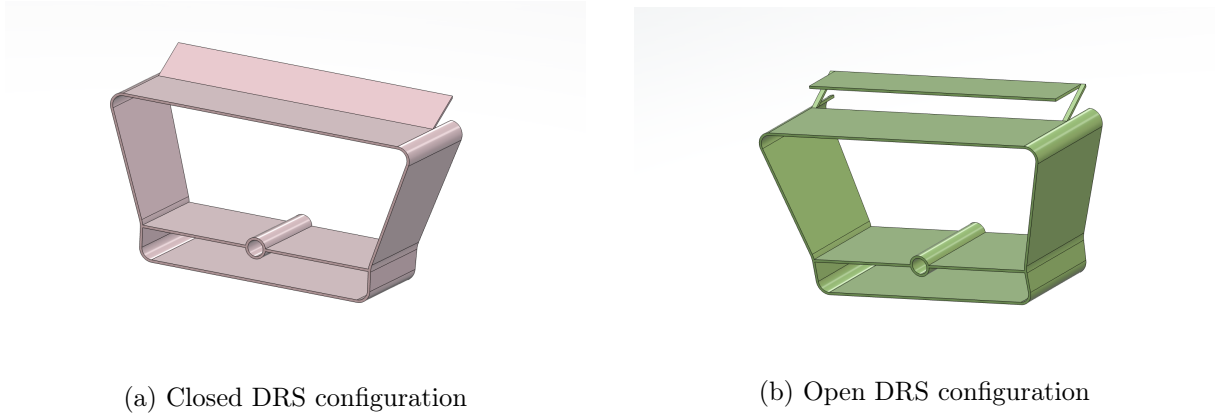


Figure 2: Rear wing geometries for the two DRS configurations.

2.6 Geometry Preparation for CFD

As OpenFOAM is the software being used for the simulations, the process is a bit less intuitive than with softwares like ANSYS Fluent for instance. In order to load the geometry, the corresponding .stl file has to be put in the folder `constant/triSurface`.

3. CFD Model and Physical Assumptions

3.1 Flow Model

Air was modelled as a Newtonian fluid with a kinematic viscosity of $\nu = 1.0 \cdot 10^{-5} m^2/s$.

3.2 Fluid Properties

The air density used in this study has a value of $\rho = 1.225 kg/m^3$

3.3 Flow Velocity

The flow velocity for this case is unidirectional with the streamwise direction being aligned on the positive x-direction. The velocity magnitude is 50 m/s.

3.4 Turbulence Model

The turbulence model used here is the k- ϵ model, and the simulation type is RAS (Reynolds-Averaged simulation).

3.5 Near-Wall Treatment

In the vicinity of the spoiler, wall functions have been used to model the flow behaviour, as can be seen on table 1.

3.6 Steady-State Assumption

Only steady state simulations were performed here.

4. Computational Domain and Boundary Conditions

4.1 Computational Domain

In order to obtain the computational domain, the file "blockMeshDict" located in the folder "system" has to be used. It is crucial to put the "scale 0.001" at the start of the file, as a conversion has to be done, between the units used in OpenFOAM and the spoiler model created in 3DEXperience. The coordinates of the volume can be put in the section "vertices", and the number of elements in the x,y, and z directions can be put in the section "blocks".

The computational domain had overall dimensions of $4.4 \text{ m} \times 3.9 \text{ m} \times 3.3 \text{ m}$ in the streamwise (X), lateral (Y), and vertical (Z) directions, respectively. The inlet and outlet were located at $x=-1.3 \text{ m}$ and $x=3.1 \text{ m}$, with the airflow oriented along the positive X-direction. A picture of this computational domain can be found on figure 3

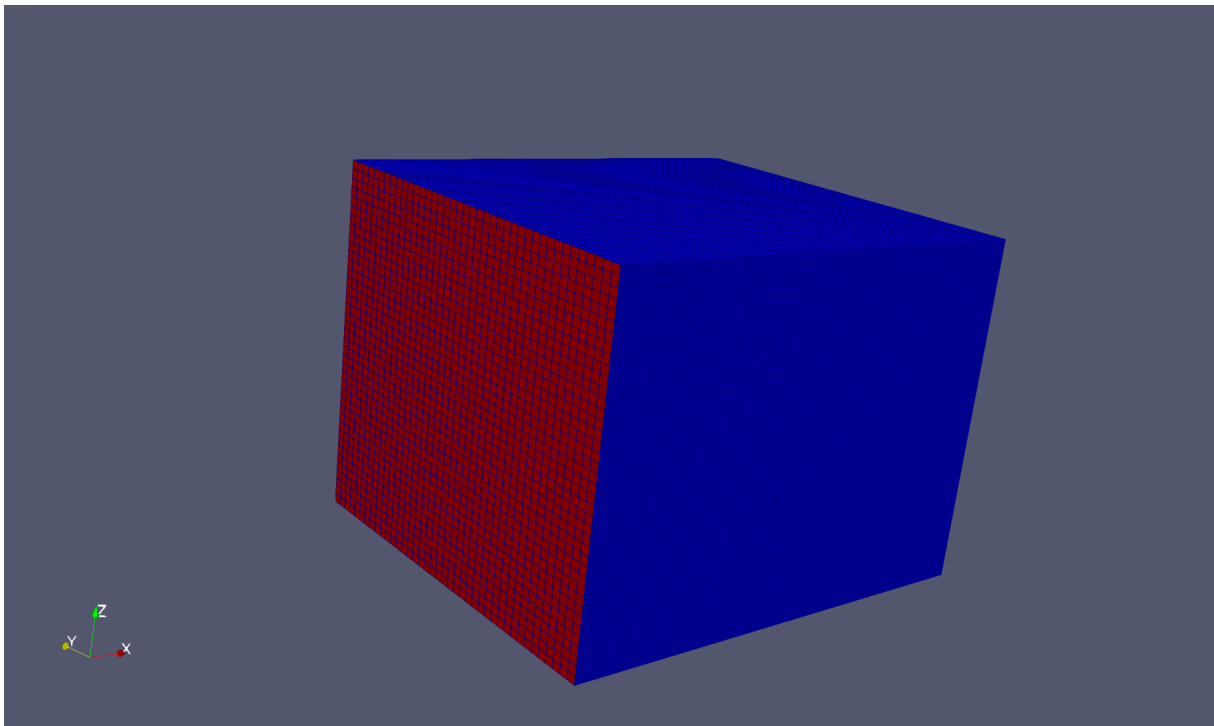


Figure 3: Computational domain

4.2 Coordinate System and Flow Direction

The orthogonal coordinate system X-Y-Z has been used, with the airflow oriented along the positive X-direction, as stated previously.

4.3 Boundary Conditions

In OpenFoam, in order to modify the boundary conditions, the file "0" has to be modified. 4 types of geometry entries can be found:

- myGeometry
- inlet
- outlet
- sidewalls

Additionnally, seven parameters have to be tuned, namely (according to their respective file names):

- epsilon
- k
- nut
- nuTilda
- omega
- p
- U

The associated parameters alongside with the related entries can be found on table 1. For both DRS configurations, the same boundary conditions were used.

Table 1: Boundary conditions used

	internalField	myGeometry	inlet	outlet	sidewalls
epsilon	uniform 14.855	type epsilonWallFunction value uniform 14.855	type fixedValue value uniform 14.855	type zeroGradient	type symmetry
k	uniform 0.375	type kqRWallFunction value uniform 0.375	type fixedValue value uniform 0.375	type zeroGradient	type symmetry
nut	uniform 0	type nutkWallFunction value uniform 0	type calculated value uniform 0	type calculated value uniform 0	type symmetry
nuTilda	uniform 0	type zeroGradient	type fixedValue value uniform 0	type zeroGradient	type symmetry
omega	uniform 440.15	type zeroGradient	type fixedValue value \$internalField	type zeroGradient	type symmetry
p	uniform 0	type zeroGradient	type zeroGradient	type fixedValue value uniform 0	type symmetry
U	uniform (0 0 0)	type fixedValue value uniform (0 0 0)	type fixedValue value uniform (50 0 0)	type zeroGradient	type symmetry

5. Mesh Generation and Discretization

5.1 Meshing Method

In OpenFoam, a specific process has to be followed in order to generate the mesh, namely:

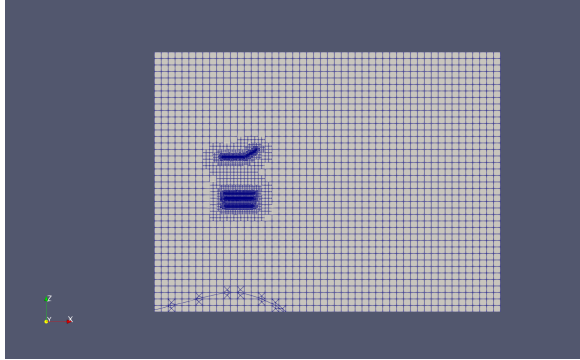
- in the file system/blockMeshDict, the external volume (computational volume) is defined through lines of command.

For this computational volume, it has been decided to go with the following numbers of elements:

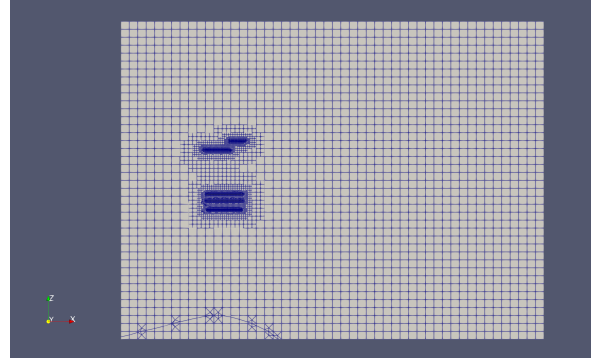
- 50 elements in the x-direction
- 40 elements in the y-direction
- 40 elements in the z-direction
- in the file system/snappyHexmeshDict, lines of command are used to define the way the mesh should be generated around the studied object (the spoiler here). Parameters such as the refinement level and the number of cell layers to model the boundary layer are defined here.

Hexahedra are the elements chosen here, as the computational volume is a very simple 3D volume .

On the figures below, pictures of the different meshes can be found. On pictures 4 and 5, a clip of the mesh in the x-z plane can be seen, with the picture 4 showing a global view of the mesh, and the picture 5 showing a detailed view of the mesh. The mesh refinement near the spoiler surface can be noticed.

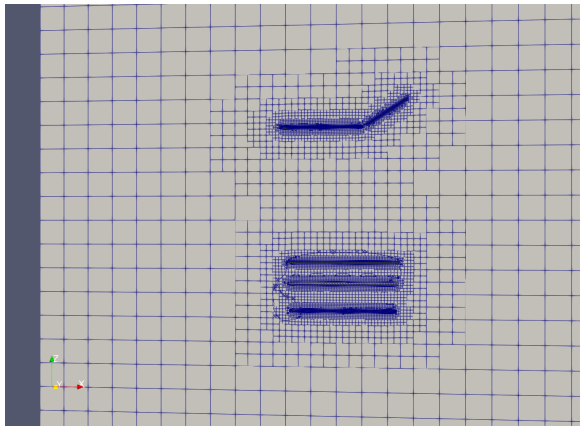


(a) Closed DRS configuration

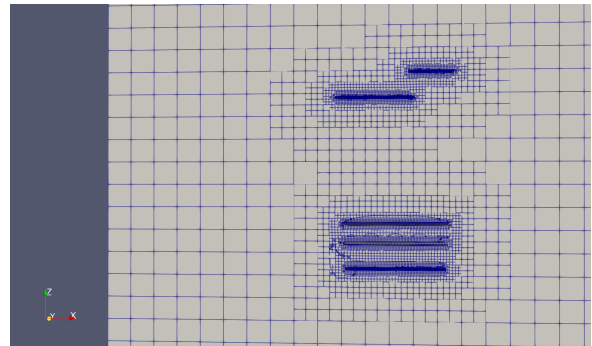


(b) Open DRS configuration

Figure 4: Computational meshes for the closed and open DRS configurations.



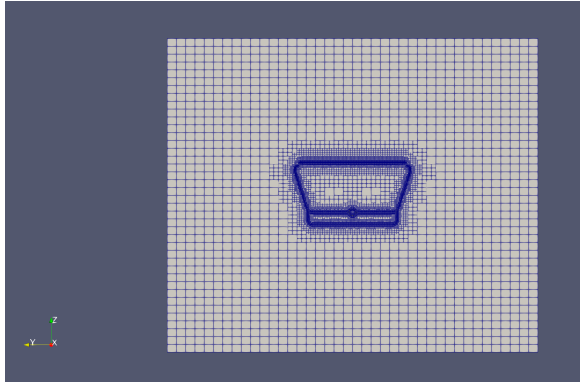
(a) Closed DRS configuration



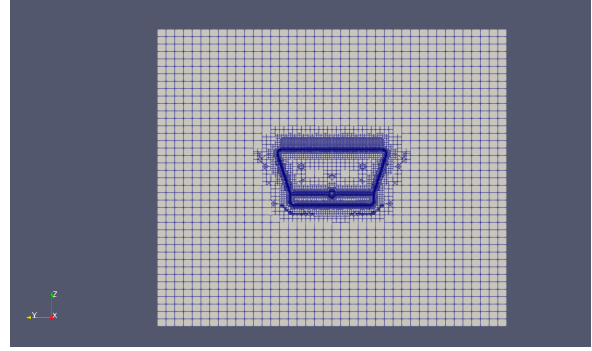
(b) Open DRS configuration

Figure 5: Detailed computational meshes for the closed and open DRS configurations.

On pictures 6 and 7, a clip of the mesh in the y-z plane can be seen, with the picture 6 showing a global view of the mesh, and the picture 7 showing a detailed view of the mesh. The mesh refinement near the spoiler surface can be noticed as well.

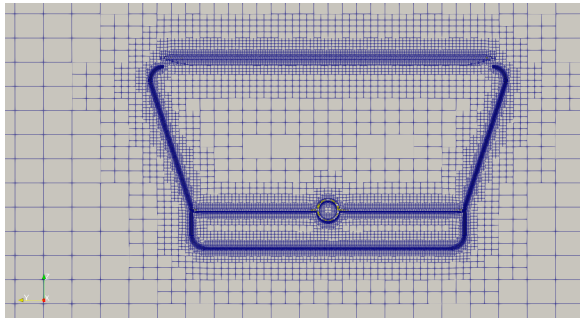


(a) Closed DRS configuration

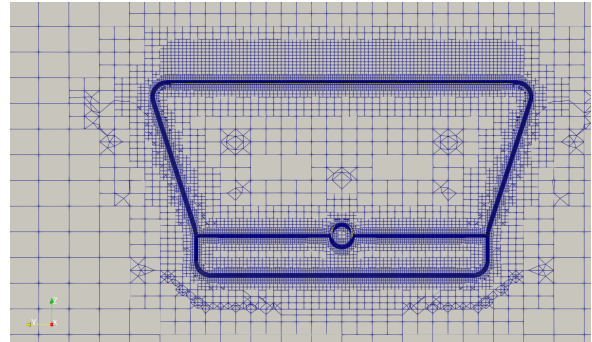


(b) Open DRS configuration

Figure 6: Close-up views of the computational meshes for the closed and open DRS configurations.



(a) Closed DRS configuration



(b) Open DRS configuration

Figure 7: Near-wall mesh details for the closed and open DRS configurations.

5.2 Surface Refinement

In order to capture the flow behaviour the best way possible near the spoiler surface, the mesh in its vicinity had to be refined in a specific way, with many layers of thin cells. The parameters that were used to enable this can be found on table 2. It can be noticed that the parameters are the same almost everywhere between the closed and open DRS configurations, except for three of them:

- expansionRatio
- finalLayerThickness
- minThickness

The reason behind these modifications with the open configuration was to improve the mesh quality, and more precisely to decrease the skewness of some surfaces.

Table 2: SnappyHexmeshDict params

	DRSclosed 50	DRSopen50
level	(3 4)	(3 4)
nSurfaceLayers	3	3
expansionRatio	1.2	1.15
finalLayerThickness	0.3	0.15
minThickness	0.1	0.05
nGrow	0	0
featureAngle	60	60
slipFeatureAngle	30	30
nRelaxIter	5	5
nSmoothSurfaceNormals	1	1
nSmoothNormals	3	3
nSmoothThickness	10	10
maxFaceThicknessRatio	0.5	0.5
maxThicknessToMedialRatio	0.3	0.3
minMedialAxisAngle	90	90
nBufferCellsNoExtrude	0	0
nLayerIter	50	50
mergeTolerance	1E-06	1E-06

5.3 Cell Count and mesh quality metrics

On table 3, some elements regarding the meshes for both configurations are analysed. It can be found that the final meshes contain mainly hexahedra elements, but also some prisms and polyhedra elements. It can be noticed as well that a warning appeared regarding the max skewness for the DRS open mesh, and that one mesh check failed as well. However, we decided to proceed with the study, as these problematic cells were tiny in volume and were likely located in the vicinity of the spoiler, where cells are elongated.

Table 3: Meshes characteristics

	DRSclosed 50	DRSopen50
number of elements	914887	894371
hexahedra	837557	821835
prisms	17866	12106
polyhedra	59464	60430
max aspect ratio	17,06	34.89
mesh non orthogonality max	59,41	59,35
average	8.09	8.02
max skewness	3,77	4.32 (warning)
Mesh check	ok	Failed 1

5.4 Near-Wall Resolution and y^+ Assessment

In order to model in the best way possible the flow behaviour at the direct vicinity of the spoiler, acceptable values of the y^+ parameters have to be obtained on the spoiler, and wall functions were used, namely for the parameters epsilon, k, and nut.

6. Numerical Setup

6.1 CFD Solver

The SIMPLE solver on OpenFOAM was used. General information regarding OpenFOAM can be found in [3].

6.2 Pressure–Velocity Coupling

The pressure-velocity coupling was handled using the SIMPLE algorithm, as implemented in the steady-state simpleFoam solver.

6.3 Spatial Discretization

For all the CFD simulations, since we used OpenFOAM, the Finite Volume Method (FVM) is used.

6.4 Convergence Criteria

The convergence criteria that have been used here are related to the residuals on U_x , U_y , U_z , p , ϵ , k . On figure 8, the residuals at the end of the simulation for both configurations are displayed. The convergence criteria that were used for each of the values of interest can be found in the file "fvSolution", in the section "SIMPLE", subsection "residualControl". The convergence criteria for the different parameters are the following:

- 10^{-2} for p
- 10^{-3} for U (U_x , U_y , U_z), ϵ and k

```
Time = 314
smoothSolver: Solving for Ux, Initial residual = 0.000140396, Final residual =
7.57215e-06, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 0.000998801, Final residual =
4.42007e-05, No Iterations 1
smoothSolver: Solving for Uz, Initial residual = 0.000774226, Final residual =
3.76679e-05, No Iterations 1
GAMG: Solving for p, Initial residual = 0.000626154, Final residual = 2.58329e-
05, No Iterations 2
GAMG: Solving for p, Initial residual = 7.0361e-05, Final residual = 5.53981e-0
6, No Iterations 2
time step continuity errors : sum local = 3.20329e-05, global = -1.43911e-06, cu
mulative = -0.0324133
smoothSolver: Solving for epsilon, Initial residual = 3.07051e-05, Final residu
al = 6.63103e-07, No Iterations 1
smoothSolver: Solving for k, Initial residual = 6.91947e-05, Final residual = 1
.67571e-06, No Iterations 1
ExecutionTime = 602.77 s ClockTime = 606 s

SIMPLE solution converged in 314 iterations
```

(a) DRS closed residuals

```
Time = 287
smoothSolver: Solving for Ux, Initial residual = 3.31698e-05, Final residual =
1.79448e-06, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 0.000971555, Final residual =
3.578e-05, No Iterations 1
smoothSolver: Solving for Uz, Initial residual = 0.000604085, Final residual =
2.72585e-05, No Iterations 1
GAMG: Solving for p, Initial residual = 0.000187179, Final residual = 1.74338e-
05, No Iterations 1
GAMG: Solving for p, Initial residual = 2.31884e-05, Final residual = 1.78483e-
06, No Iterations 3
time step continuity errors : sum local = 5.75897e-06, global = 8.76703e-08, cum
ulative = -0.0395207
smoothSolver: Solving for epsilon, Initial residual = 1.02316e-05, Final residu
al = 2.28634e-07, No Iterations 1
smoothSolver: Solving for k, Initial residual = 4.03636e-05, Final residual = 9
.56625e-07, No Iterations 1
ExecutionTime = 565.98 s ClockTime = 568 s

SIMPLE solution converged in 287 iterations
```

(b) DRS open residuals

Figure 8: Final residuals for the closed and open DRS configurations.

6.5 Aerodynamic Force Calculation

In order to compute the aerodynamic forces, the file "controlDict" had to be modified. In the "functions" section, some lines had to be added. These lines can be found on picture 9. It can be seen that the air density has to be input, as well as the reference object from which the forces must be computed (here, the spoiler).

```

functions
{
    forces
    {
        type          forces;
        libs          ("libforces.so");

        patches       (myGeometry);

        rho           rhoInf;
        rhoInf        1.225;

        CofR          (0 0 0);

        writeControl   timeStep;
        writeInterval  1;
    }
}

```

Figure 9: Functions section in controlDict file

7. Results

7.1 Flow Field

On picture 10a , the flow field around the spoiler for the DRS closed configuration is displayed. It can be noticed that after the spoiler, the flow is deflected upwards. This deflection is linked to the downforce applied on the car, that sticks to the ground to improve its stability.

On picture 10b , the flow field around the spoiler for the DRS open configuration is displayed. It can be noticed that after the spoiler, the flow is not deflected this time. It translates into a higher speed for the car, due to less drag, but to a reduced stability.

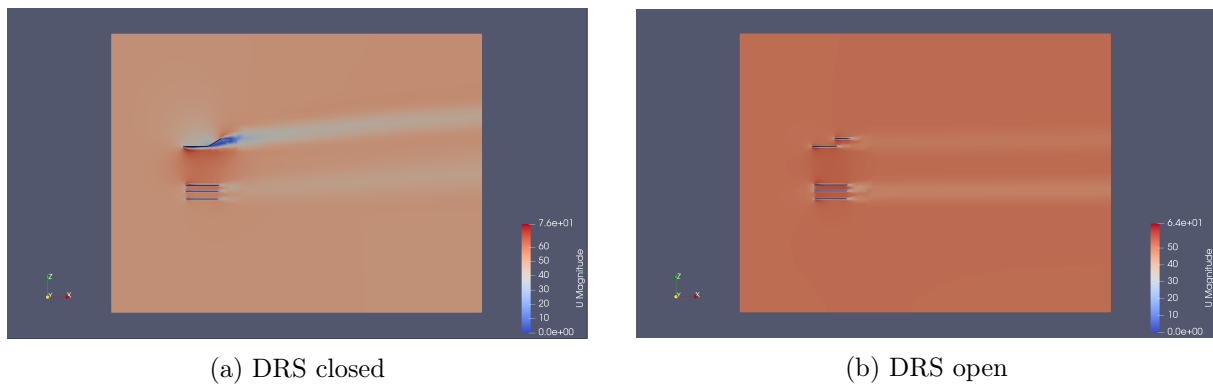


Figure 10: Velocity magnitude for the closed and open DRS configurations at 50 m/s.

7.2 Pressure Distribution

The picture distribution for the closed configuration is shown on picture 11a. It can be noticed that there is a pressure that is exerted on top of the spoiler, which is responsible for the resulting downforce. The particular shape of the spoiler explains this pressure increase.

On picture 11b, the pressure distribution around the spoiler with the open DRS configuration is displayed. The stagnation point at the leading edge of each of the four blades can clearly be seen, and are related to a local pressure increase. The pressure drop between the spoiler blades that was noticeable on the closed condition can be seen in this configuration as well. This is due to Bernoulli's effect, where an increase in speed translates in a pressure decrease. However, this applies only for laminar flows, so not in the direct vicinity of the spoiler.

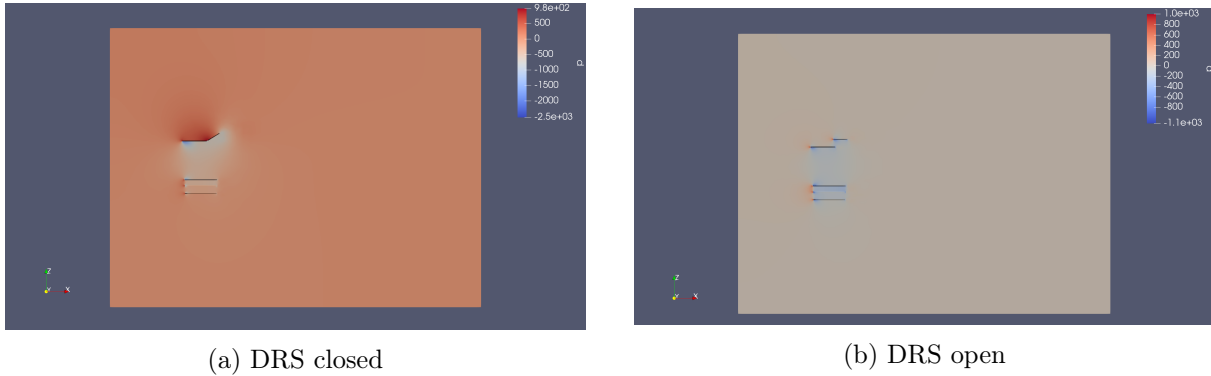


Figure 11: Pressure distribution for the closed and open DRS configurations at 50 m/s.

7.3 Aerodynamic Forces

The different aerodynamic forces exerted on the spoiler were computed, and their values at the last iterations are provided on figure 12, where a screenshot of the console has been taken for each of the two configurations. The main results are displayed on table 4, where it can be noticed that from the closed configuration to the open configuration, the drag force goes from 239,65 N to 99,94 N, and the downforce goes from 619,035 N to 86,6 N. The minus sign used for the downforce values indicates that the force is directed downwards.

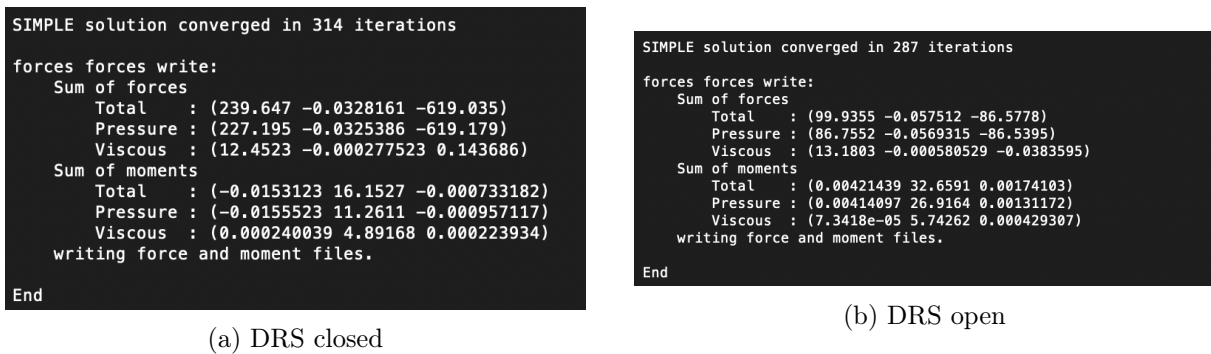


Figure 12: Aerodynamic force results for the closed and open DRS configurations at 50 m/s.

Table 4: Aerodynamic forces applied on the spoiler

	F_x (N)	F_z (N)
Closed configuration	239,647	-619,035
Open configuration	99,94	-86,6

7.4 y^+ Distribution

On figure 13, the y^+ distribution around the spoiler is shown for both cases. As can be seen, the average y^+ values were approximately 135 for both configurations, which is consistent with the high-Reynolds-number wall-function approach adopted in this study. However, significant local variations remained, particularly near geometrical edges and highly curved regions.

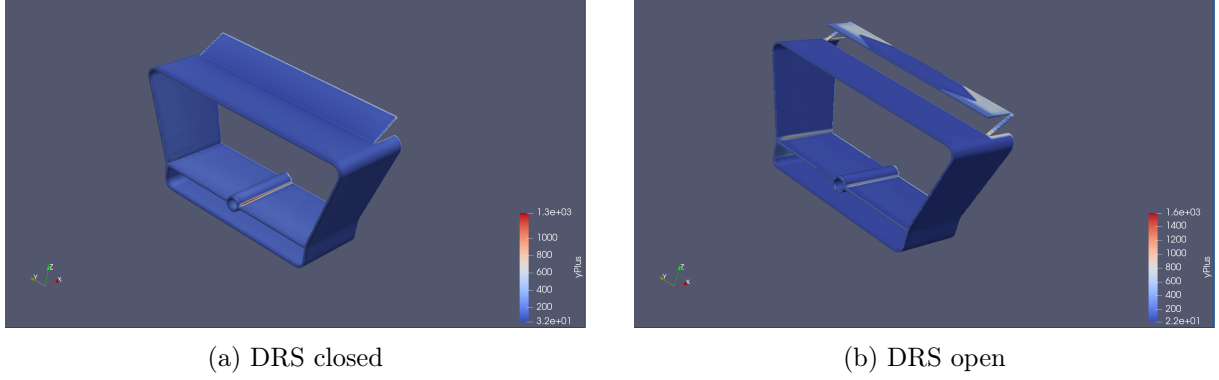


Figure 13: y^+ distribution for the closed and open DRS configurations at 50 m/s.

Table 5: Reference y^+ values for both configurations

Configuration	y^+_{min}	y^+_{avg}	y^+_{max}
DRS closed	15,57	135,04	1876,8
DRS open	4,20	133,89	1743,1

8. Comparison of DRS Configurations

8.1 Drag Comparison

When opening the DRS, the drag force drastically decreases, by about 58,3 %.

8.2 Downforce Comparison

When opening the DRS, the downforce decreases even further, by about 86,1 %.

8.3 Drag Coefficient

8.3.1 Reference Area Definition

The reference area definition will play a crucial role regarding the drag coefficient computation. By definition, it is the projected area of the spoiler, orthogonal to the flow direction.

8.3.2 Drag Coefficient Comparison

The drag coefficient C_d was computed thanks to the following equation:

$$C_d = \frac{F_d}{0,5 \cdot A \cdot \rho \cdot U^2} \quad (1)$$

with:

- C_d being the drag coefficient

- F_d being the drag force
- A being the spoiler reference area
- ρ being the air density
- U being the wind speed

Two different methods were used:

- In the first method, the frontal area of the spoiler with the closed DRS configuration was taken as a reference, for the C_d value computations of both configurations
- In the second method, the frontal area used is respective to each of the two configurations.

Table 6: Drag coefficient computations with both methods

	DRS closed	DRS open
1st method	1,06	0,44
2nd method	1,06	1,11

The results are displayed on table 6. It can be noticed that the C_d values for the DRS closed configurations are the same for both methods. This makes sense, as the same area was used. However, it can be noticed that the C_d values for the open DRS configuration drastically change from the first to the second method. This can be explained by the fact that the projected area of the DRS device is non negligible compared to the whole spoiler frontal area. At least but not last, it is not intuitive that opening the DRS would increase the drag coefficient of the spoiler. However, it is explained by the fact that the projected spoiler area between the two configurations changes a lot. Thus it was decided to proceed with the values computed using the 1st method.

9. Discussion

9.1 Drag–Downforce Trade-Off

Both the drag force and the downforce decreased dramatically when opening the DRS. For a drag force decrease of about 58,3 %, the downforce decreases by about 86,1 %.

9.2 Influence of Near-Wall Mesh Refinement

The near-wall mesh refinement allowed to capture in a better way the flow behaviour in the vicinity of the spoiler surface, and thus, to model the boundary layer in a more accurate way. This increase in accuracy translated in a reduced value of the average y^+ distribution around the spoiler. For the closed DRS configuration, before and after the near-wall mesh refinement, the average y^+ value went from approximately 600 to approximately 135.

9.3 Limitations of the CFD Model

The geometry is really simplified compared to a real car, and the number of boundary layers could be increased. Moreover, the steady-state hypothesis is somewhat a simpler way to model the flow around the spoiler. There could also be work to be done to make the y^+ value distribution around the spoiler more homogenous.

9.4 Recommendations for Future Improvements

Here are some recommendations for future improvements:

- Increase the number of elements in order to further increase the model accuracy
- Improve the detail level of the spoiler, to be even closer from reality
- Discard the steady state hypothesis and use transient simulations instead
- Further improve the cell layers analysis around the spoiler to make the y^+ value distribution around the spoiler more homogeneous

10. Conclusion

This project allowed us to quantify a well known effect in formula 1 racing: opening the DRS allows the car to go faster, as the drag force is reduced. However, it also decreases the stability of the car, as the downforce is also decreased. As found, for a drag force decrease of about 58,3 %, the downforce decreases by about 86,1 %. The objective for speed and performance optimization is to find the best tradeoff between the smallest decrease of the downforce force on one hand to maximize the car stability, and the biggest decrease in drag force to maximize the car speed.

References

- [1] *Ferrari SF24 Formula 1 F1 Blueprints*. Available at: https://www.the-blueprints.com/vectordrawings/show/28902/ferrari_sf24_formula_1_f1/ (Accessed July 2026).
- [2] *Ferrari SF24 Rear Wing Details*. Available at: <https://fr.motorsport.com/f1/news/ferrari-aileron-arriere-2023-djeddah/10584165/> (Accessed July 2026).
- [3] *OpenFOAM Documentation*. Available at: <https://www.openfoam.com/documentation/overview> (Accessed August 2026).